

1.		5
1.1.		5
1.2.		5
2.		6
2.1.		6
2.2.		6
2.3.		6
2.4.		7
2.5.		7
2.6.		7
2.7.		7
2.8.		8
3.		9
3.1.		9
3.2.		9
3.3.		9
3.4.		9
3.5.		10
3.6.		10
3.7.		10
3.8.		10
3.9.		11
3.10.		11
3.11.		11
3.12.		11
3.13.		12

3.14.		12
3.15.		12
3.16.		13
3.17.		13
3.18.		13
3.19.		14
3.20.		14
3.21.		15
3.22.		15
3.23. n		15
4.		16
4.1.		16
4.2.		16
4.3.		16
4.4.		16
4.5.		17
4.6.		17
4.7.		17
4.8.		18
4.9.		18
4.10.		18
4.11.		19
5.		20
5.1.		20
5.2.		20
5.3.		20
5.4.		20
5.5.		21
5.6.		21
5.7.		21
5.8.		22

6.		23
6.1.		23
6.2.		23
6.3.		23
6.4.		23
6.5.		23
6.6.		24
6.7.		25
6.8.		25
6.9.		25
6.10.		26
6.11.		26
6.12.		27
6.13.		27
6.14.		27
6.15.		27
7.		29
7.1.		29
7.2.		29
7.3.		30
7.4.		30
7.5.		30
7.6.		30
7.7.		31
7.8.		31
7.9.		31
7.10.		31
7.11.		32
7.12.		32
7.13.	2	32
7.14.		33

7.15. 33

1.

1.1.

2022

pp. 40 - 41

1.2.

2022

pp. 43, 47 - 54

12

$X \text{ cm}^2$

$Y \text{ g}$

(1)

$$Y = aX + b$$

(2)

(3)

$$(X, Y) = (11, 23), (23, 49), (5, 10), (50, 115), (32, 64), (21, 45) \\ (16, 35), (39, 84), (25, 50), (45, 100), (5, 11), (34, 70)$$

2.1.

(1)

2.2.

- (1)
- (2)
- (3)
- (4) 3

- (1) c
- (2) $x = 1, 2, 3, 4, 5, 6$ $P(X = x)$
- (3) X $E[X]$
- (4) $P(X \geq 4)$

2.4.

2023

pp. 81, 86

$$f(x) = \begin{cases} \frac{c}{x^3} & (x \geq 10) \\ 0 & (x < 10) \end{cases}$$

(1) c

(2)

- (4)
- (5)
- (6)1

2.8.

2025						
(1)	X	μ	σ^2			
(2)	200		60	6		42
	78					

3.

3.1.

2024	p. 107		
6	4	1	2
(1)	2		
(2)	2		

3.2.

2026	p. 107		
	3	5	
(1)	1	1	
	(a)	2	
	(b)	2	
	(c)	2	1
(2)			2
	1		
(3)	(1)	(2)	

3.3.

2022	pp. 107 - 108		
1	2	3	4
			2
X_1	X_2	$Y = X_1 - X_2 $	
(1)	Y		
(2)	Y		
(3)	Y		

3.4.

2023	2025	pp. 113 - 114
(X, Y)		

$$f(x,y)=\begin{cases}6e^{-2x-3y} & (x\geq 0,y\geq 0) \\ 0 & (\text{otherwise})\end{cases}$$

$X \quad Y$

3.5.

2026

$X \quad Y$

$$\int_0^\infty e^{-x}dx=1,\qquad \int_0^\infty xe^{-x}dx=1$$

(1) $f(x,y)=\begin{cases}xe^{-x-y} & (x\geq 0,y\geq 0) \\ 0 & (\text{otherwise})\end{cases}$

(a) $f_X(x) \quad f_Y(y)$

(b) $X \quad Y$

(2) $f(x,y)=\begin{cases}c(x+y)e^{-x-y} & (x\geq 0,y\geq 0) \\ 0 & (\text{otherwise})\end{cases}$

(a) c

(b) $X \quad Y$

3.6.

2022

pp. 103 - 104, 113

1 4 2 2 1

$X=0$ $X=1$ 2

$Y=0$ $Y=1$

(1) (X,Y)

(2) $X \quad Y$

3.7.

2022

pp. 67, 114 - 115

2 $A \quad B$

3.8.

2022

2023

2026

p. 115

3

$\frac{1}{2}$

(1)

1

2

(2)

1

2

A

2

3

B

A

B

(3)

$P(A|B)$

3.9.

2024

p. 115

2

A

4

B

2

7

C

2

9

(1)

A

B

(2)

A

C

3.10.

2023

pp. 72 - 73, 109 - 110, 114 - 115

2

E

F

$P(E) = a$

$P(F) = b$

(1)

$P(E \cup F)$

a

b

(2)

$P(E \cap F^c | E \cup F)$

a

b

F^c

F

3.11.

2025

1

5

X

Y

$Z = X - Y$

(1)

X

Y

(2)

Z

(3)

Z

(4)

Z

3.12.

2025

pp. 118 - 119

(1)	2	A	B						
(2)		500	1	X				X	
	2	K_1	K_2			3 : 1			
			K_1, K_2			0.98	0.90		
						0.01			
				K_1					

3.13.

2022	2023		pp. 118 - 119						
	3	5			1				
	2								
(1)	1								
(2)	2		1						

3.14.

2023		pp. 118 - 119							
	1	10%			3				
(1)									
(2)		2							

3.15.

2022		pp. 118 - 119							
	5	4	3						
1		100 g	45 g	30 g					
(1)		1							
		40 g							

(2)

2

100 g

2

(3)

3

150 g

3

3.16.

2022

2024

pp. 118 - 119

A

0.1%

A

/

99%

10%

(1)

(2)

A

1

3.17.

2023

pp. 118 - 119

99%

99%

0.5%

1

- (1)
- (2)
- (3)
- (4)

3.18.

2023

pp. 118 - 119

S

S

K

- S

K

99%

1%
- S

K

99%

1%

(1)

K

S

(2)

(1)

3.19.

2025

pp. 118 - 119

A

B

- (1) $P(A|B) = 0.98$ $P(A|B^c) = 0.005$ $P(B) = 0.001$
- (2) $P(A|B^c) = 0.003$ $P(B) = 0.0005$ $P(B|A) \geq 0.142$ $P(A|B)$

3.20.

pp. 118 - 119

500

1

XXX

2

K_1

K_2

$K_1 : K_2 = 3 : 1$

K_1

0.98

K_2

0.90

0.01

(1)

K_1

(2)

(1)

3.21.

2023

pp. 122 - 124

p

$0 < p < 1$

X

Y

$$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$$

$$Z = XY$$

(1)

$E[Z]$

(2)

X

Z

$E[(X - E[X])(Z - E[Z])]$

(3)

X

Z

p

p

Y

Z

(4)

(3)

p

$\Pr[X + Y + Z \leq 2]$

3.22.

2023

2024

pp. 125 - 126

2

X

Y

(1)

a

(2)

X

Y

$\text{Cov}(X, Y)$

(3)

X

Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

3.23. n

2023

p. 127

n

1

n

$(n + 1)$

4.

4.1.

2025

2 $B(2,p)$

4.2.

2024

p. 141

1 6

(1) 10 5
(2) 100 25

4.3.

2023

2026

pp. 93 - 95, 141

p n
 X 2 $\text{Bin}(n,p)$ $q = \left(\frac{X}{n}\right)$
(1) $P(|q - p| \leq 0.02) \geq 0.95$ n

(2) (1) n (2)
(3) p
(3) (1) -
 n
(4) (2) (3) n

4.4.

2024

pp. 143 - 145, 153

(1) :

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$$y = f(x) \qquad (x,y)$$

(2)
$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

(3)

4.5.

2022 pp. 121 - 122, 143

(1) $X, Y \sim N(170, 6^2)$

(a) $W = \frac{X + Y}{2}$

(b) X, Y, W

(2) X, Y

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \quad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$$W = XY$$

4.6.

2024 pp. 147 - 151

(1) $X \sim N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2)
$$\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \right)$$

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01 \quad z_0$

(c) $z(0.005)$

4.7.

2024 pp. 149 - 151

(1) $a, Z \sim N(0, 1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2)

X

$N(m, \sigma^2)$

$$P\left(|X - m| \geq \frac{\sigma}{4}\right)$$

(3)

m

σ

n

$$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$$

$$n$$

4.8.

2024

2026

pp. 149 - 150

$$E$$

$$N(66.31, (23.55)^2)$$

(1)

$P(X \geq 80)$

(2)

(3-a)

(3)

80

95

(4)

20%

4.9.

2025

pp. 149 - 150

$$C$$

(1)

95%

C

(2)

4.10.

2022

pp. 153 - 154

- (1) 6
- (2) 10000
- (3) 1000049005100

4.11.

pp. 93 - 95, 153 - 154

200606

X

(1) $42 \leq X \leq 78$

(2)

(3) $XN(60, 6^2)P(42 \leq X \leq 78)$

5.

5.1.

2023

2025

pp. 168 - 169

2

5.2.

2024

pp. 93, 168 - 169

1

ATM

ATM

1

40

ATM

1

60

ATM

/

60

75

90

105

120

5.3.

2023

pp. 166 - 169

(1)

3

5

(2)

1

2

1

4

5.4.

2025

4

(1)

1

3

2

(2)

X

$P(X \leq 3)$

$P(X = 3) = 5P(X = 5)$

5.5.

2023

pp. 166 - 169

1%

200

(1)

X

(2)

(3)

3

(2)

(3)

5.6.

2024

pp. 153 - 154, 166 - 169

4

(1)

240

6

38

45

(2)

0.1%

500

3

5.7.

2026

$p = 0.001$

$n = 500$

X

$e^{-0.5} \approx 0.6065,$

$\sqrt{0.5} \approx 0.7071$

(1)

(a)

X

(b)

$E[X]$

$\text{Var}(X)$

(2)

(a)

(b)

$P(X \geq 3)$

(3)

(a)

(b)

$$P(X \geq 3)$$

(4)

(a)

(2)

(3)

(b)

$$np$$

5.8.

2022

2023

pp. 139 - 143, 168 - 169

p

$0 < p < 1$

$(1 - p)$

N

$N > 1$

(1) $N = 5$

(2) $t \quad 0 \leq t \leq N$

(3) $m \quad m =$

N

$0 \leq m \leq N$

m

(4) $n \quad 0 \leq n \leq N \quad N - n \quad P(n\&N, p)$

(5) (4) n

(6) $n \quad \lambda = Np \quad N \rightarrow \infty \quad (4) \quad P(n\&N, p)$

$$P(n; \lambda) \quad \lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$$

6.

6.1.

2022p. 174

2

- (1)
- (2)

6.2.

2025

μ σ^2 n $X_1, X_2, ..., X_n$.

(1) $X_1, X_2, ..., X_n$

(2).

(3) n $S_n = X_1 + X_2 + \cdots + X_n$.

(4)

6.3.

2025p. 179

118020

(1) 251170

(2) N 0.91175

N

6.4.

2025

1.2 kg0.1 kg

100122.5 kg

6.5.

2022pp. 202 - 205

μ

σ^2

3

$X_1 \quad X_2 \quad X_3$

(1)

$.$

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \frac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \frac{X_1 + X_2 + X_3}{3}$

μ

(2)

$\{T_1, T_2, T_3\}$

6.6.

2023

2023

2025

pp. 176 - 180, 202 - 205

(1)

0

1.2

3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4)

θ

$\Theta_n = T(X_1, X_2, ..., X_n)$

3

(a)

(b)

(c)

(5)

3

6.7.

20222024p. 205

100010

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

6.8.

2025pp. 179, 216

p n X

(1) X .

(2) n X

(3) p_0 95% n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq$
114)

6.9.

2022pp. 180 - 182

p p 10,000

$3,500$ $p = \frac{3500}{10000} = 0.35$

(1) p

(2) p 2

6.10.

2025

- (1)

$q = \frac{Y}{n}$

A

p

A

Y

$100(1 - \alpha)\%$
- (2)

S

TV

T

140

32

T
- (a)

p

95%
- (b)

p

3%

0.95
- (c)

6.11.

2024

2024

pp. 223 - 225

- (1)

$Bin(1, p)$

$X_1, X_2, ..., X_n$
- (a)
- (b)
- (2)

n

A

Y

$q = \frac{Y}{n}$

A

p

$100(1 - \alpha)\%$
- (a)
- (b)
- (3)

S

TV

T

140

32

T
- (a)

p

95%
- (b)

p

3%

0.95
- (c)

(2)

$$\frac{N}{\sigma^2}$$

$$n$$

(3)

$$/$$

7.

- ① 仮説の検定
- ② 統計量の検定
- ③ 優位水準と棄却域の設定
- ④ 帰無仮説の棄却・選択

7.1.

2022

2023

pp. 227 - 231

(1)

(2)

(3)

(4)

5%

(3)

5%

(5)

(1)

(1)

(6)

(6)

(7)

• (8:)

• (9:)

7.2.

2022

pp. 227 - 234

(1)

(2)

7.3.

2023	pp. 2235 - 236				
		8000			
	25			7700	
			5%		
		640			

7.4.

2023	pp. 238 - 240				
	T				
S		365			6
		330, 350, 385, 321, 340, 365			
	5%				

7.5.

2025	pp. 238 - 240				
	T				
S		360			6
		350, 355, 385, 390, 370, 365			

- (1)10%
- (2)5%

7.6.

2023	2024	2024	2025	pp. 235 - 236
(1)	μ		$N(\mu, 49)$	49
	$H_0 : \mu = 4,$	$H_1 : \mu > 4$		
	m			
	(a)	0.05	$m = 6$	2

(b)0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

20.20.1

7.7.

2022pp. 241 - 242

(1) 2A B 2

1012

A2.040.03B

2.020.022

5%

2

(2) A B

5%

7.8.

2022pp. 241 - 242

10070

6815

5%

7.9.

2024pp. 252 - 253

31

(1) A 10A

31A

5%

(2) 10731

7.10.

2024pp. 252 - 253

5

1

5%

7.11.

2023

pp. 252 - 253

10

X

p

$H_0 : p = \frac{1}{2}$

H_0

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$ 5%

(b) $H_1 : p \neq \frac{1}{2}$ 2%

7.12.

2023

pp. 252 - 253

5

1

p

(1) 550

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3
 $p = \frac{1}{2}$

(b) $p = \frac{3}{4}$ 3

7.13. 2

256

116

5%

(1) 2

.

(2) 2 10%

7.14.

20232025pp. 252 - 253

$H_0 : p = \frac{1}{2}$

$0 < p < 1$

H_0

H_0

333

(1) 1 1

(2) 2 2

(3) 2 $\frac{1}{3}$ p

7.15.

2024pp. 252 - 253

A B C A 3 B 2 C 1

1

(1) A A $\frac{1}{16}$ B B $\frac{1}{9}$ C C $\frac{1}{25}$

11

(2) 960 640

(1) 5%